MARYLEBONE ROSTER · LINKS DESIGNER

Proposed CEA Link — December 2026

How this rota would work from December 2026, and how it compares with today's

Prepared 28 September 2026 · every figure is worked out from the rota itself by the Marylebone Roster app, not typed in

PROPOSAL

Targeted Fatigue Redo

A cover week at line 18, with the weeks ordered to reduce fatigue

CODE	**TF-24-EXT**
FINGERPRINT	**8eef9a13**
FAMILY	Weekday Lates
FIRST CREATED	22 Sep 2026

✓ **24** lines, one per person · ✓ **35h 00m** a week — exactly the contract · ✓ **0** hard-limit breaches · ✓ **0** rests under 12h · ⚠ **4** fatigue factors present — 3 as rostered (today: 5)

Five questions a manager asks first, in order. Every figure is worked out from the rota; **page 2 says how to read each one**, with today's link and the other proposals marked on a scale.

CAN IT BE RUN?

Yes

the hard limits

0 rests under 12h · longest run 9 of 13 · contract exact

DOES IT MEET THE DECEMBER RULES?

3 of 11

rules met

the staffing levels, the ticket office, the floor and the handovers — page 7 · today's link meets 3 of 11

HOW TIRING IS IT?

4

fatigue factors present, of 25

advisory — present means worth a look, not a breach · today 5 · fewest of the others 0

DOES IT FOLLOW THE TRAINS?

37.2

how closely the floor follows the weekday trains — 0 is a perfect match

today 68.8 · Saturday 19 · Sunday 67.9 · best of the others 20.4

IS IT FAMILIAR?

13 of 19

shift times people work today

6 new, listed on page 5 · 4 of 20 working weeks keep one shift time (today 7 of 16)

5 in 24

full weekends off

21% of the rotation, against 20% today (4 in 20) · most of the others 25%

4 cover weeks

lines 1, 7, 12, 18

not evenly spaced — gaps of 7, 6, 5, 6 lines · today's sit at 1, 7, 12, 17

4h 10m a week

typical week-to-week move

how far the working day shifts each week — today's link moves 4h 0m a week

What this is

Targeted Fatigue Redo is a proposal for the CEA link on the December 2026 timetable: a 24-line rotation for 24 people, where today's link has 20. In short: a cover week at line 18, with the weeks ordered to reduce fatigue.

Against today's link — the table below; page 2 explains each figure.

AT A GLANCE	TODAY'S LINK	TARGETED FATIGUE REDO
December 2026 rules met	3 of 11	**3 of 11**
People on duty — weekday · Saturday · Sunday	11–12 · 10 · 8	**13–16 · 12 · 9**
How closely the floor follows the trains — weekday · Sat · Sun (lower is closer)	68.8 · 68.7 · 72.5	**37.2 · 19 · 67.9**
ORR fatigue factors present (of 25)	5	**4**
Most days worked in a row (limit 13)	9	**9**
Shortest rest between two duties (at least 12h)	12h 30m	**13h 35m**
Full weekends off	4 in 20	**5 in 24**
Different shift times (how many already worked today)	18	**19 (13)**

What it is not. Not a recommendation and not a finished roster: nothing here has been through the roster office or a rep yet. The December 2026 staffing levels were passed on for this exercise and have no document behind them yet.

HOW TO READ THE NUMBERS

What each figure means, and what good looks like

Every figure on this sheet is worked out from the rota. On each scale: this sheet in navy, today's 20-line link as a ring, the best and worst of the other 22 proposals as gold and grey ticks.

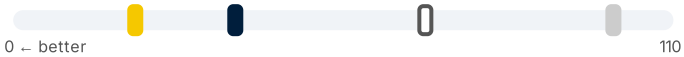
Three things to keep in mind. Lower is better on the fit and the fatigue count; higher is better on rest, weekends, weeks on one shift time and familiarity. “Present” is not “failed” — the ORR list reports what is there. Today's link is the yardstick, not the target: it meets 3 of the 11 December 2026 rules.

DOES IT FOLLOW THE TRAINS?

How well the floor follows the trains — weekday

For each hour of an average weekday, the share of the day’s floor staff on duty is set against the share of the day’s trains: an hour that carries a tenth of the trains should have about a tenth of the staff. The score adds up how far each hour is off, so 0 would be a perfect match. In plain terms a typical hour here is about 1.4 percentage points off the trains, against 2.0 today. The ticket office is left out of both — it is staffed to its opening hours, not to the trains.

What it does not tell you. It compares shares, not numbers: the same extra person in every hour changes nothing, and an extra person in a quiet hour makes the score worse even though nobody is worse off. Read it beside the hour-by-hour numbers on page 6.



worst of the others **100** · best of the others (Right Away) **20.4**
today’s link **68.8** · this sheet **37.2**

Saturday **19** (today 68.7) · Sunday **67.9** (today 72.5). Everyone on duty, ticket office included: weekday 34.4 (today 51.1).

On page 6.

HOW TIRING IS IT?

Fatigue factors present

How many of the 25 roster patterns on the Office of Rail and Road’s good-practice list appear anywhere in the rotation — long runs of earlies, short gaps between duties, start times that jump — counted by the Links designer’s own module.

What it does not tell you. A factor present is worth a look, never a breach: the ORR says the list is guidance, not limits. Zero does not mean approved and one does not mean unsafe; page 8 names which, and how big.

worst of the others **7** · best of the others (By the Book, Eight Forty and 5 more) **0**
today’s link **5** · this sheet **4**

On page 8.

CAN IT BE RUN?

Longest run of worked days

The most consecutive days anyone works in the worst case — a cover week’s four duties placed as badly as they can be. Chiltern’s limit is 13; the app’s own target is 6.

What it does not tell you. The roster clerk decides where cover-week duties fall, so the figure as actually rostered can be lower; page 8 shows both readings.

worst of the others **9** · best of the others (By the Book, Eight Forty and 11 more) **6**
today’s link **9** · this sheet **9**

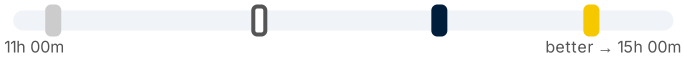
On page 7.

CAN IT BE RUN?

Tightest rest between duties

The shortest gap anywhere between one duty ending and the next beginning — Saturday into Sunday and the last line into the first included. Twelve hours is the floor; a design under it cannot be run.

What it does not tell you. It is one gap. A rotation with many 12h05 rests and one with none read the same here; the count under 12h is in the chips on page 1.



worst of the others **11h 15m**
best of the others (Anchored Lines, Frozen Block) **14h 30m**
today’s link **12h 30m** · this sheet **13h 35m**

On page 7.

WHAT IS IT LIKE TO WORK?

Full weekends off

The share of weeks with both Saturday and Sunday off. Today’s link gives 4 in 20, one weekend in 5; this sheet gives 5 in 24, about one in 5.

What it does not tell you. A weekend off beside a 42-hour week is still a weekend off; the hours by week are on the grid, page 4.

worst of the others **8%**
best of the others (Anchored Lines, By the Book and 11 more) **25%**
today’s link **20%** · this sheet **21%**

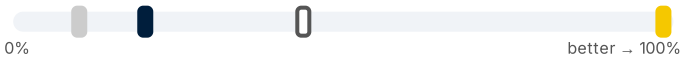
On page 5.

WHAT IS IT LIKE TO WORK?

Working weeks on one shift time

The share of working weeks where a person keeps the same clock time all week. A week that mixes an early and a late is what people mean by “all over the place”.

What it does not tell you. One shift time all week can still be a hard week if the shift is long; the week-to-week move and the duty lengths are on pages 5 and 8.



worst of the others **10%** · best of the others (Weekend Capped) **100%**
today’s link **44%** · this sheet **20%**

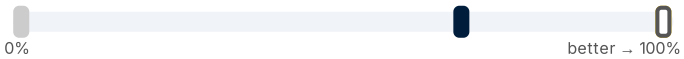
On page 5.

IS IT FAMILIAR?

Shift times people work today

The share of this sheet’s clock times that somebody on the 20-line link already works. It is a rough guide to how much there is to learn and get used to.

What it does not tell you. Familiar is not the same as popular — a time everyone works and nobody likes still counts — and a new time in one week is not the same as one in ten. The duty table on page 5 lists the new ones.



worst of the others **0%** · best of the others (Same Turns) **100%**
today’s link **100%** · this sheet **68%**

On page 5.

BEFORE THE FIGURES

The decision this sheet supports

What is being decided, which criteria are fixed and which are yours to weigh, how much each figure can bear, and what no page here can tell you. Read this before the figures.

1 What this is
2 How to read the numbers
3 The decision it supports (this page)
4 The 24-line grid, with the words explained
5 A week compared with today
6 People on duty, hour by hour
7 Hard limits and the December 2026 figures
8 The ORR fatigue factors
9 Beside the other proposals
10 The grid, ready to paste into the app

The question

Should the CEA link for the December 2026 timetable be **Targeted Fatigue Redo**? It is judged against today's 20-line link and the December 2026 rules, and page 9 sets it beside the other 22 proposals on the same figures.

Targeted Fatigue Redo — a cover week at line 18, with the weeks ordered to reduce fatigue.

The criteria, in three tiers

HARD — A DESIGN MEETS THESE OR CANNOT BE RUN

- ✓ Twelve hours between duties — met, tightest 13h 35m
- ✓ Chiltern's 13-day limit — longest run 9 days
- ✓ The 35-hour contract, exactly — 35h 00m

SOFT — WEIGH THESE AGAINST EACH OTHER

- December 2026 rules met: **3 of 11** (today 3 of 11)
- Fatigue factors present: **4** worst case, 3 as rostered (today 5)
- How the floor follows the trains, weekday · Sat · Sun: **37.2 · 19 · 67.9** (today 68.8 · 68.7 · 72.5; lower is closer)
- Times worked today: **13 of 19** · full weekends off: **5** in 24 (today 4 in 20)

PREFERENCE — FOR STAFF TO SAY, NOT THE APP

- Early against late: earlies 7h 15m–8h 30m, lates 7h 30m–9h 10m
- The 6 new times: 07:00-15:30, 08:00-16:00, 08:30-17:00, 10:30-19:00, 12:00-20:30, 16:25-23:55
- The longest duty, 9h 10m, and who holds it
- Cover weeks at lines 1, 7, 12, 18

How much to trust each figure

FIGURE ON THESE PAGES	WHERE IT COMES FROM	WEIGHT
0 rests under 12h · 35h 00m contract · 12/9 at the weekend	Worked out from the rota; checked again by the app when it is loaded	**Firm**
Longest run 9 days, inside 13	Computed; the written source of the 13-day limit is still to be confirmed	**Firm figure**
At least four at the open, three at the close, five at 22:00, 14 and 10 at the weekend	Relayed December 2026 headcounts with no document behind them	**Provisional**
4 fatigue factors present	The ORR's 2021 list, which it states are not limits; 4 factors whose exact definition is still to be confirmed	**Advisory only** — never pass or fail
Demand fit 34.4 · 20.7 · 62.4; on the floor 37.2 · 19 · 67.9	The measured December 2026 timetable; one way of measuring, not the only one; floor takes the ticket office out of both sides	**A guide**
13 of 19 times are today's	Computed against the live 20-line link	**Firm**, but only a rough guide to what staff will accept

What no page here can tell you

- Whether the people who would work them accept the times nobody works today (07:00-15:30, 08:00-16:00, 08:30-17:00, 10:30-19:00, 12:00-20:30, 16:25-23:55) — this sheet counts times already worked today, which is only a rough guide to what staff will accept.
- How the roster clerk places the four duties of a cover week. It can move the 48-hour-break figure (12 if the four are worked together, 15 if they are split up), and the rota itself does not decide it.
- How leave and sickness cover land on a 24-line link with four cover weeks (lines 1, 7, 12, 18); nothing here models absence.
- Whether the December 2026 timetable demand curve holds on Wembley event days — the curve is a measured timetable, not a footfall count.

Before it is agreed

Lines 2, 6, 9, 16, 17 hold the times nobody works today; the people who would work those weeks read them first and say in one line each what they would change. The roster office answers the cover-week question. Someone confirms the source of the 13-day limit. Until then, this sheet is an aid to discussion, not a decision.

THE ROTATION

The 24-line link

Sunday to Saturday per line; everyone moves down one line each week and line 24 goes back to line 1. Hours and days at the right, cover beneath.

LINE	SUN	MON	TUE	WED	THU	FRI	SAT	MON–SAT hours	ALL WEEK hours	DAYS WORKED Mon–Sat	TURNS Mon–Fri
1	SP	SP	SP	SP	SP	SP	SP	*35h 00m*	*35h 00m*	—	—
2	14:30 23:25	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	RD	RD	30h 00m	38h 55m	4	**1**
3	07:15 15:45	RD	08:30 16:30	06:20 14:20	06:20 14:20	06:20 14:20	06:20 14:20	40h 00m	48h 30m	5	2
4	RD	RD	06:20 14:20	08:30 16:30	RD	13:30 22:00	14:45 23:55	33h 40m	33h 40m	4	3
5	14:30 23:25	RD	14:00 22:30	14:00 22:30	14:00 22:30	16:25 23:55	RD	33h 00m	41h 55m	4	2
6	RD	06:20 14:20	07:00 15:30	07:00 15:30	07:00 15:30	08:30 17:00	RD	42h 00m	42h 00m	5	3
7	SP	SP	SP	SP	SP	SP	SP	*35h 00m*	*35h 00m*	—	—
8	07:15 15:45	06:20 13:45	RD	RD	13:30 22:00	14:00 22:30	13:30 21:00	31h 55m	40h 25m	4	3
9	RD	12:00 20:30	12:00 20:30	12:00 20:30	12:00 20:30	12:00 20:30	RD	42h 30m	42h 30m	5	**1**
10	RD	14:00 22:30	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	14:45 23:55	47h 40m	47h 40m	6	2
11	RD	07:00 15:30	07:00 15:30	RD	RD	06:20 13:35	06:20 14:50	32h 45m	32h 45m	4	2
12	SP	SP	SP	SP	SP	SP	SP	*35h 00m*	*35h 00m*	—	—
13	RD	RD	14:00 22:30	13:30 22:00	14:00 22:30	14:00 22:30	14:30 22:00	41h 30m	41h 30m	5	2
14	RD	16:25 23:55	16:25 23:55	RD	RD	07:00 15:30	06:20 14:50	32h 00m	32h 00m	4	2
15	07:15 15:45	06:20 13:45	06:20 13:35	06:20 13:35	06:20 13:35	07:00 15:30	RD	37h 40m	46h 10m	5	3
16	RD	08:30 17:00	08:30 17:00	08:30 17:00	RD	RD	RD	25h 30m	25h 30m	3	**1**
17	10:30 19:00	16:25 23:55	RD	RD	06:20 14:20	06:20 14:20	06:20 14:50	32h 00m	40h 30m	4	2
18	SP	SP	SP	SP	SP	SP	SP	*35h 00m*	*35h 00m*	—	—
19	14:30 23:25	RD	06:20 13:45	06:20 13:45	06:20 13:45	06:20 13:45	08:00 16:00	37h 40m	46h 35m	5	1
20	08:30 16:30	06:20 13:45	RD	RD	13:30 22:00	16:25 23:55	13:30 22:00	31h 55m	39h 55m	4	3
21	13:30 22:00	13:30 22:00	RD	13:30 22:00	13:30 22:00	13:00 21:00	RD	33h 30m	42h 00m	4	2
22	RD	14:00 22:30	13:30 22:00	13:30 22:00	13:30 22:00	RD	RD	34h 00m	34h 00m	4	2
23	RD	RD	RD	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	30h 00m	30h 00m	4	**1**
24	RD	RD	06:20 13:45	06:20 13:45	07:00 15:30	RD	06:20 13:45	30h 45m	30h 45m	4	2
Cover	**9** SP 4	**13** SP 4	**15** SP 4	**15** SP 4	**16** SP 4	**16** SP 4	**12** SP 4	35h 00m	38h 13m	4.35	4 one turn
Minutes	4,635	6,225	7,205	7,205	7,780	7,690	5,895	**42,000**	**46,635**		

Early turns · Late turns · Cover (spare) week — four duties of seven, any turn · Rest day

Mon–Sat hours average 35h 00m over 24 lines, a cover week counted as a contracted week · days worked average over the 20 working lines · Turns: how many different shift times the week works Monday to Friday; **bold** means one shift time all week · Minutes: total duty minutes each day — Monday to Saturday adds up to the contract (20 × 35h = 42,000)

The words on this page

Link — the whole rota: 24 weeks laid out as 24 lines. Everyone works line 1, then line 2, and so on round the wheel; the link needs 24 people.

Turn — a shift time, such as 06:20-13:45. An **early** starts before 09:00; a **late** starts after.

Cover (spare) week — a week with no fixed times. You work four duties of the seven days, placed by the roster clerk to cover leave and sickness.

The contract — 35 hours a week, averaged over the 24 lines; a cover week counts as a full contracted week.

Line — one week of the pattern, Sunday to Saturday. You move down a line each week, and line 24 goes back to line 1.

Rest day (RD) — a day off inside the contracted week. Two together make a proper break; one on its own does not count as a 48-hour break.

Sunday — not part of the contract. A Sunday duty is rest-day working, paid on top of the week, which is why the hours are shown Mon–Sat and all week.

One-turn week — the same clock time all week (shown bold in the Turns column); the easiest kind of week to live around.

AGAINST TODAY'S LINK

A week, and the duty table

How a working week compares with today's, and every shift time beside the ones worked today.

The shape of a week, against today

MEASURE	TODAY (20)	PROPOSED (24)
Weeks on one shift time	7 of 16	**4 of 20**
Weeks mixing early and late shifts	7	**6**
Rest-day breaks that are two days or more	13 of 17	**15 of 25**
Single rest days between duties (not a 48h break)	4	**10**
Days worked in a week (weeks × days)	1×3, 3×4, 11×5, 1×6	**1×3, 6×4, 9×5, 4×6**
Cover (spare) weeks (whole weeks)	lines 1, 7, 12, 17	**lines 1, 7, 12, 18**
Different shift times (6 not worked today)	18	**19**
Mon–Sat hours, average week (the contract)	35h 00m	**35h 00m**
Sunday duties (not contracted; half the working lines)	8 of 16 lines	**9 of 20 lines**

Changed — how many people are on duty

PEOPLE ON DUTY	TODAY	PROPOSED	WHY
Monday to Friday	11–12	**13–16**	20 working weeks of 35 hours have to be worked somewhere
Saturday	10	**12**	the December 2026 figure
Sunday	8	**9**	the December 2026 figure
Ticket office wk · Sat · Sun	4 · 4 · 3	**4 · 4 · 4**	not floor cover; Sunday one late today
Opening at 06:20 (07:15 Sunday) wk · Sat · Sun	4 · 4 · 3	**4 · 5 · 3**	at least four to open, every day — not met: Sunday 3
Through to the close (23:55; 23:25 Sunday) wk · Sat · Sun	2 · 2 · 3	**3–4 · 3 · 3**	at least three to close, every day — met
Still on duty at 22:00 wk · Sat · Sun	4 · 2 · 3	**4–6 · 3 · 3**	at least five from 22:00, every day — not met: Wed 4, Saturday 3, Sunday 3

The duty table, beside today's

TURN	WK	SAT	SUN	WK	SAT	SUN
06:20 - 13:35 7h 15m · Early	1			**1**		
06:20 - 13:45 7h 25m · Early	1			**3** +2	**1** +1	
~~06:20 - 14:00 7h 40m · Early~~		~~1~~			~~-1~~	
06:20 - 14:20 8h 00m · Early	2			**2**	**1** +1	
06:20 - 14:50 8h 30m · Early		3			**3**	
07:00 - 15:30 8h 30m · Early				**2** +2		
07:15 - 15:45 8h 30m · Early			3			**3**
08:00 - 16:00 8h 00m · Early					**1** +1	
~~08:00 - 16:30 8h 30m · Early~~	~~2~~			~~-2~~		
08:30 - 16:30 8h 00m · Early			1	**1** +1		**1**
08:30 - 17:00 8h 30m · Early				**1** +1		
10:30 - 19:00 8h 30m · Early						**1** +1
~~11:00 - 19:30 8h 30m · Late~~	~~1~~			~~-1~~		
~~12:00 - 20:00 8h 00m · Late~~		~~1~~			~~-1~~	
12:00 - 20:30 8h 30m · Late				**1** +1		
13:00 - 21:00 8h 00m · Late			1	**1** +1		-1
13:30 - 21:00 7h 30m · Late		1			**1**	
13:30 - 22:00 8h 30m · Late	1			**4** +3	**1** +1	**1** +1
14:00 - 22:30 8h 30m · Late	2			**2**		
14:30 - 22:00 7h 30m · Late		2			**1** -1	
14:30 - 23:25 8h 55m · Late			3			**3**
14:45 - 23:55 9h 10m · Late		2			**2**	
~~15:15 - 23:55 8h 40m · Late~~	~~2~~			~~-2~~		
16:25 - 23:55 7h 30m · Late				**4** +4	**1** +1	

Busiest weekday · Saturday · Sunday. The first three columns are what the 20-line link does now; the last three are the proposal. Green cells grew, amber shrank, a struck-through row is a time the proposal does not use.

A weekday adds up to 10,590 minutes of duty and a Saturday 5,895, so five weekdays and a Saturday make exactly 20 × 35 hours = 42,000; Sunday adds up to 4,635, paid on top of the contracted week. 6 of the 19 times are ones nobody works today.

COVER AGAINST THE SERVICE

People on duty, hour by hour

Cover today and proposed against the measured December 2026 timetable (arrivals and departures, weighted by train length). A darker orange hour carries more of the day's trains.

How to read it: each blue cell is the number of people on duty in that hour (darker blue = more people). The orange row is the timetable — how much of the day's trains that hour carries (darker orange = busier). Read the columns top to bottom: cover should be thickest where the orange is darkest. The **fit** figure at the right end of each row scores that match for the whole day.

Monday to Friday window 06:20–23:55 · a row for each different weekday

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY MON · WED		4	4	6	6	6	7	7	7	7	7	7	5	5	5	4	4	4	2	55.8
TODAY TUE · THU · FRI		4	4	6	6	6	7	7	8	8	8	8	6	6	6	5	5	4	2	50
IN THE TICKET OFFICE		2	2	2	2	2	2	2	2	4	2	2	2	2	2	2	2	2		
ON THE FLOOR MON · WED		2	2	4	4	4	5	5	5	3	5	5	3	3	3	2	2	2	2	83.7
ON THE FLOOR TUE · THU · FRI		2	2	4	4	4	5	5	6	4	6	6	4	4	4	3	3	2	2	64.4
PROPOSED MON		4	5	6	6	6	6	7	8	7	6	8	7	7	7	7	6	5	3	28.4
PROPOSED TUE		4	6	8	8	8	8	9	10	9	8	9	7	7	7	7	6	5	3	43.6
PROPOSED WED		4	5	7	7	7	7	8	11	9	8	10	8	8	8	8	7	4	3	42.3
PROPOSED THU		4	6	6	6	6	6	7	11	11	9	10	10	10	10	10	9	5	3	48.8
PROPOSED FRI		4	6	7	7	7	7	8	10	10	8	10	9	9	9	9	7	6	4	31
IN THE TICKET OFFICE		2	2	2	2	2	2	2	2	4	2	2	2	2	2	2	2	2		
ON THE FLOOR MON		2	3	4	4	4	4	5	6	3	4	6	5	5	5	5	4	3	3	31.1
ON THE FLOOR TUE		2	4	6	6	6	6	7	8	5	6	7	5	5	5	5	4	3	3	50
ON THE FLOOR WED		2	3	5	5	5	5	6	9	5	6	8	6	6	6	6	5	2	3	48.7
ON THE FLOOR THU		2	4	4	4	4	4	5	9	7	7	8	8	8	8	8	7	3	3	62.3
ON THE FLOOR FRI		2	4	5	5	5	5	6	8	6	6	8	7	7	7	7	5	4	4	35
TRAINS, DEC 2026	5	69	107	127	122	95	85	75	84	79	87	105	140	138	107	90	73	75	66	

Saturday window 06:20–23:55

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY		4	4	4	4	4	4	5	6	9	6	6	6	6	6	5	4	2	2	45.7
IN THE TICKET OFFICE		2	2	2	2	2	2	2	2	4	2	2	2	2	2	2	2			
ON THE FLOOR		2	2	2	2	2	2	3	4	5	4	4	4	4	4	3	2	2	2	68.7
PROPOSED		5	5	6	6	6	6	6	8	10	6	6	6	6	6	6	5	3	3	20.7
IN THE TICKET OFFICE		2	2	2	2	2	2	2	2	4	2	2	2	2	2	2	2			
ON THE FLOOR		3	3	4	4	4	4	4	6	6	4	4	4	4	4	4	3	3	3	19
TRAINS, DEC 2026	6	24	58	75	81	92	88	77	64	68	72	84	82	67	64	79	78	62	41	

Sunday window 07:15–23:25

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY			3	4	4	4	4	4	5	8	8	5	4	4	4	4	3	3	3	65.9
IN THE TICKET OFFICE			2	2	2	2	2	2	2	3	3	1	1	1	1	1	1	1		
ON THE FLOOR			1	2	2	2	2	2	3	5	5	4	3	3	3	3	2	2	3	72.5
PROPOSED			3	4	4	5	5	5	6	9	9	6	5	5	4	4	4	3	3	62.4
IN THE TICKET OFFICE			2	2	2	2	2	2	4	4	4	2	2	2	2	2	2	2		
ON THE FLOOR			1	2	2	3	3	3	2	5	5	4	3	3	2	2	2	1	3	67.9
TRAINS, DEC 2026			15	45	65	71	74	71	61	55	64	69	78	73	75	52	71	65	50	

Ticket office. Not floor cover, and staffed today exactly as in the plan Monday to Saturday (two early, two late). **Sunday differs:** today two earlies and **one** late — a 14:30–23:25 closer, in the office until 22:30 — where the plan has two and two. This design does not roster the office as pairs, so its floor is everyone on duty less the plan's posts. The indented rows take it out of each side, so the floor rows compare like with like.

Cover weeks are not in these figures — a cover week has no fixed times — so real cover is a little higher. **Fit** is one number for how closely the people on duty follow the trains through the day: 0 would be a perfect match, lower is better. The fit of the average weekday is the one on page 1. Sunday: 5 December 2026 timetable movements fall after the 23:25 finish — the standing question on whether Sunday's window moves; the window is stored per design, so this proposal can be rebuilt to either answer.

Reading it. The **floor** rows are the ones to read: the ticket office is staffed to its opening hours, not to the trains, so every everyone-on-duty row fits more loosely than its floor row. The fit measures **shape, not numbers** — each hour's share of the day's cover against its share of the day's trains — so an extra person in a quiet hour makes it worse even though nobody is worse off. Read it beside the numbers of people, never alone.

THE RULES IT IS ASSESSED AGAINST

The checks sheet

Hard limits first — met or not. Then the ORR good-practice fatigue factors, which report what is present and never pass or fail a design.

Hard limits — a design either meets these or cannot be run

✓ **More than 13 consecutive days worked** — longest possible run **9** days (today: 9)

Longest possible run is 9 days, within Chiltern's limit of 13. A cover week is 4 duties in 7 days and the roster clerk places them, so this is the worst case the link allows. Worked as a block of four rather than split day-on-day-off, the answer is the same — **9** days — so this row does not depend on how a cover week is placed.
The written source of the 13-day limit is still to be confirmed.

✓ **At least 12 hours between duties** — **0** rests under 12h anywhere in the rotation, Saturday-into-Sunday and line-into-line included (today: 0)

Tightest anywhere: **13h 35m — line 20 Fri 16:25-23:55 into Sat 13:30-22:00** (today's tightest: 12h 30m).

✓ **The contracted week, exactly** — **35h 00m** average Mon–Sat over the 24 lines, cover weeks counted as contracted weeks

700h of duty a week across 20 working lines. Sunday duties (77.25h in all) are paid on top as rest-day working, as they are today. Individual weeks range 25h 30m–47h 40m; only the average is the contract.

The December 2026 rules — the same rules every proposal is measured against

RULE	PROPOSAL	
At least four on at the open, every day	**✕ 4 · 5 · 3**	06:20; 07:15 on a Sunday
At least three through to the close, every day	✓ 3–4 · 3 · 3	23:55; 23:25 on a Sunday
At least five still on duty at 22:00, every day	**✕ 4–6 · 3 · 3**	a 22:00 finish is not "on at 22:00"
Fourteen on a Saturday, ten on a Sunday	**✕ 12 · 9**	
Four cover weeks, evenly spread	**✕ lines 1, 7, 12, 18**	whole weeks, placed by the roster clerk
The ticket office: two early and two late, identical turns, every day	**✕ Mon–Fri no · Sat no · Sun no**	Mon–Fri 06:20–14:20 and 14:00–22:30; Sat 06:20–14:50 and 14:30–22:00; Sun from 07:15 and to 22:30
Every weekday closer starts at 15:45	**✕ 16:25-23:55**	
At least two on the floor at every moment	**✕ fewest 2 · 3 · 1**	the ticket office left out; checked every five minutes
Handovers: 15 minutes to each closer, 20 in the ticket office	✓ floor yes · office 20 min	on a Sunday every opener also stays 15 minutes past the last closer arrives
Sunday duties between 8h and 9h	✓ 8h00–8h55	
No more shift times than today	**✕ 19 against today's 18**	

Still to settle

- **8 of the 11 December 2026 rules are not met** — at least four on at the open, every day; at least five still on duty at 22:00, every day; fourteen on a Saturday, ten on a Sunday; four cover weeks, evenly spread; the ticket office: two early and two late, identical turns, every day; every weekday closer starts at 15:45; at least two on the floor at every moment; no more shift times than today. The shaded rows above give the figures.
- **New shift times.** 6 of the 19 are times nobody works today; page 5 lists them.
- **Sunday's finish.** Five December 2026 timetable movements fall after the 23:25 close; whether Sunday's window moves is not decided here.

THE RULES IT IS ASSESSED AGAINST · CONTINUED

ORR fatigue factors

Good practice guidelines — Fatigue Factors, p3 (December 2021). ⚠ present — the pattern is in this design and worth a look, not a breach · ✓ clear — it is not · ● standing — true of the station itself, not of any design · – does not apply here. This is the list the link is assessed against.

The Office of Rail and Road lists 25 roster patterns that tend to tire people — long runs of earlies, short gaps between duties, start times that jump about. For each one this table asks whether the pattern is in the rotation, today and proposed, and how big it is. The list is guidance: a factor present is a question for the room, never a pass or a fail, and a design showing nothing is not thereby approved.

CODE	FACTOR	TODAY'S LINK	PROPOSED
FF1	Night shift covering 00:00–05:00 Time of day	– –	– –
FF2	Early shift starting 05:00–07:00 Time of day	● 24	● 25
FF3	Very early shift starting before 05:00 Time of day	✓ 0	✓ 0
FF4	Very early shift (before 05:00) over 8h Duty length	– –	– –
FF5	Day shift over 12h Duty length	✓ 0	✓ 0
FF7	Early shift over 10h Duty length	✓ 0	✓ 0
FF8b	Fewer than 2 days rest after a block of early starts Recovery time	⚠ 3	⚠ 5
FF11	More than 13 consecutive shifts without a 48h break Recovery time	✓ 12 ✓ 9 as rostered	⚠ 15 ✓ 12 as rostered
FF10	More than 4 consecutive 12h day shifts Cumulative	✓ 0	✓ 0
FF15	More than 4 consecutive early shifts in a rotating pattern Cumulative	⚠ 6	⚠ 7
MRSF	More than 12 consecutive day shifts Cumulative	✓ 9	✓ 9
MRSF	More than 55 hours worked in any 7-day period Cumulative	⚠ 60.8	✓ 54.7
MRSF	More than 7 consecutive 8h shifts (definition to confirm) Cumulative	⚠ 9	✓ 7
MRSF	More than 6 consecutive night or early shifts in a permanent pattern Cumulative	– –	– –
FF17	Backward rotating pattern (definition to confirm) Circadian	✓ 10 backward / 12 forward	✓ 12 backward / 17 forward
FF18	Rotating pattern of about a week (definition to confirm) Circadian	● 20-line rotation · typically 4h 0m a week	● 24-line rotation · typically 4h 10m a week
FF19	Successive start times varying by more than 2 hours (definition to confirm) Circadian	⚠ 6	⚠ 8
FF6	Night shift over 10h Night shifts	– –	– –
FF8	Less than 2 days rest after consecutive nights Night shifts	– –	– –
FF9	Less than 14h rest in any 24h for night shifts Night shifts	– –	– –
FF12	Only one day rest after night shifts Night shifts	– –	– –
FF14	More than 4 consecutive nights in a rotating pattern Night shifts	– –	– –
FF16	More than 3 consecutive night shifts over 8h Night shifts	– –	– –
FF20	First night shift Night shifts	– –	– –
FF13	Less than 12h rest in any 24h Intervals between duties	✓ 0	✓ 0

Present: 4 in the worst case, **3** as rostered (today 5) · standing: 2 · FF2 fires on every 06:20 duty, so it is a property of the station's opening time rather than of any design. **Two readings of a cover week, on the one row they move.** A cover week is worked four days of seven and the link does not say which four, so a run figure is a range. The headline number is the ceiling — the four split day-on-day-off, which supplies no 48-hour break and joins the blocks either side. It is reachable: of the 35 ways to place four duties in seven days, the 10 that leave no two rest days together produce exactly it. Worked as a block, which is what a cover week looks like on the roster, the three rest days fall together and the week always supplies a break — that is the *as rostered* figure beneath it. Checked on every row: FF11 is the only one where the two differ. Which reading applies is a question for the roster office, not for this sheet. This sheet is an aid to a conversation, not a fatigue risk assessment.

BESIDE THE OTHERS

Where it stands among the 23

Every proposal on the same figures, with today's link at the foot. This sheet's row is in bold.

Like with like. Every figure below was worked out the same way for every proposal, from its rota, by the Marylebone Roster app — so any two rows can be compared directly.

PROPOSAL	RULES MET	FATIGUE FACTORS	LONGEST RUN	FULL WEEKENDS OFF	FLOOR FIT — WK · SAT · SUN			SHIFT TIMES	WORKED TODAY
Right Away	11/11	0	6	6	20.4	11.6	19.2	17	5
Pinned Turns	8/11	0	6	6	30.9	31.7	58.1	16	9
Round Times	8/11	0	6	6	33.1	30.8	58.1	14	6
Quarter To	8/11	0	6	6	54.9	48.7	100.9	13	12
Weekend Capped	7/11	1	6	6	54.9	30.9	65.7	11	9
Same Turns	7/11	1	6	6	57.5	48.7	100.9	15	15
By the Book	5/11	0	6	6	31.9	9.7	30.9	19	0
Eight Forty	5/11	0	6	6	100	23.4	19.8	26	2
Office Written In	4/11	0	6	5	96.6	12.7	69.1	26	4
Gates Mended	4/11	1	6	6	33.3	45.4	39.3	15	4
Light Retime	4/11	3	9	4	37.2	31.1	47	23	11
Saturday Four	4/11	3	9	5	37.2	30.7	67.9	18	10
Weekday Lates	4/11	5	9	6	41.4	19	67.9	18	13
Fifteen Turns	4/11	7	9	2	33.3	47.3	39.3	15	5
Anchored Lines	3/11	1	7	6	38.1	19	67.9	19	13
Frozen Block	3/11	1	6	4	38.1	19	67.9	19	13
Evening Peak	3/11	1	6	6	41.9	19	67.9	19	13
Tenth Sunday	3/11	2	6	5	37.2	31.1	47	24	11
Cover at Seventeen	3/11	3	9	5	37.2	19	67.9	19	13
Short Closer	3/11	3	9	4	37.2	46.3	67.9	21	12
Targeted Fatigue Redo	**3/11**	**4**	**9**	**5**	**37.2**	**19**	**67.9**	**19**	**13**
Three Mondays	3/11	4	9	5	37.2	19	67.9	19	13
Weeks 17-18 Swapped	3/11	5	9	6	37.2	19	67.9	19	13
Today's 20-line link	3/11	5	9	4	68.8	68.7	72.5	18	18

Ordered by rules met, then fewest fatigue factors, then the weekday floor fit — an order, not a verdict. **Rules met** counts the 11 December 2026 rules on page 7. **Floor fit** is how closely the people on the floor, the ticket office taken out, follow the trains (lower is closer; page 6). **Longest run** is days worked in a row at worst, against Chiltern's limit of 13; **full weekends off** are out of 24 (20 for today's link).

LOAD IT

The rotation, ready to paste

Line number, then Sunday to Saturday. SP is a cover week. Paste the whole block into Links → Import.

Links → Import, then paste the whole block below. Line number, then Sunday to Saturday; SP is a cover week. The Links designer re-runs every check on these pages from the pasted cells, so nothing here has to be taken on trust.

The same rotation is also supplied beside this PDF as Targeted-Fatigue-Redo-TF-24-EXT-import.txt (tab-separated, pastes directly) and Targeted-Fatigue-Redo-TF-24-EXT.json (the app's own format).

1	SP	SP	SP	SP	SP	SP	SP
2	14:30-23:25	16:25-23:55	16:25-23:55	16:25-23:55	16:25-23:55	RD	RD
3	07:15-15:45	RD	08:30-16:30	06:20-14:20	06:20-14:20	06:20-14:20	06:20-14:20
4	RD	RD	06:20-14:20	08:30-16:30	RD	13:30-22:00	14:45-23:55
5	14:30-23:25	RD	14:00-22:30	14:00-22:30	14:00-22:30	16:25-23:55	RD
6	RD	06:20-14:20	07:00-15:30	07:00-15:30	07:00-15:30	08:30-17:00	RD
7	SP	SP	SP	SP	SP	SP	SP
8	07:15-15:45	06:20-13:45	RD	RD	13:30-22:00	14:00-22:30	13:30-21:00
9	RD	12:00-20:30	12:00-20:30	12:00-20:30	12:00-20:30	12:00-20:30	RD
10	RD	14:00-22:30	16:25-23:55	16:25-23:55	16:25-23:55	16:25-23:55	14:45-23:55
11	RD	07:00-15:30	07:00-15:30	RD	RD	06:20-13:35	06:20-14:50
12	SP	SP	SP	SP	SP	SP	SP
13	RD	RD	14:00-22:30	13:30-22:00	14:00-22:30	14:00-22:30	14:30-22:00
14	RD	16:25-23:55	16:25-23:55	RD	RD	07:00-15:30	06:20-14:50
15	07:15-15:45	06:20-13:45	06:20-13:35	06:20-13:35	06:20-13:35	07:00-15:30	RD
16	RD	08:30-17:00	08:30-17:00	08:30-17:00	RD	RD	RD
17	10:30-19:00	16:25-23:55	RD	RD	06:20-14:20	06:20-14:20	06:20-14:50
18	SP	SP	SP	SP	SP	SP	SP
19	14:30-23:25	RD	06:20-13:45	06:20-13:45	06:20-13:45	06:20-13:45	08:00-16:00
20	08:30-16:30	06:20-13:45	RD	RD	13:30-22:00	16:25-23:55	13:30-22:00
21	13:30-22:00	13:30-22:00	RD	13:30-22:00	13:30-22:00	13:00-21:00	RD
22	RD	14:00-22:30	13:30-22:00	13:30-22:00	13:30-22:00	RD	RD
23	RD	RD	RD	16:25-23:55	16:25-23:55	16:25-23:55	16:25-23:55
24	RD	RD	06:20-13:45	06:20-13:45	07:00-15:30	RD	06:20-13:45

Prepared 28 September 2026. The code **8eef9a13** in every footer is the design's fingerprint — eight characters computed from these 168 cells and nothing else. If the Links designer shows the same code after you paste, you have exactly this design; if one cell differs, the code changes. Every figure in this document was computed by the Marylebone Roster app's Links modules (v24.35) from exactly these cells; import them and the Links designer’s Design checks, hard limits, fatigue factors and coverage cards will restate them.